Selection is easy with Timing Pulleys and Belts automatic calculation tool available at:
http://fawos.misumi.jp/FA_WEB/pulley_sea/

[Step 6] Check if Inter-Shaft Distance Adjustment Range is Larger than that in Table 30

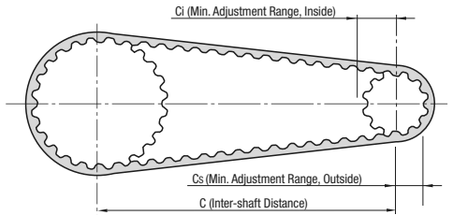


Table 31 Minimum Inter-Axial Distance Adjustment Range

Belt Length	Length Tolerance	Inter-Shaft Distance Tolerance	MXL		XL		L		H		S2M S3M S5M		S8M S14M		MTS8M		P2M P3M P5M		P8M		T5		T10	
			Ci	Cs	Ci	Cs	Ci	Cs	Ci	Cs	Ci	Cs	Ci	Cs	Ci	Cs	Ci	Cs	Ci	Cs	Ci	Cs	Ci	Cs
150 or Less	±0.35	±0.18		3		3		3		3		2		3		–		3		3		3		3
150 to 250	±0.41	±0.21		3		3		3		3		2		3		3		3		3		3		3
250 to 380	±0.46	±0.23		5		5		5		5		2		3		3		3		3		5		5
380 to 500	±0.51	±0.26		10		10		10		10		2		3		3		3		3		10		10
500 to 750	±0.60	±0.30		10		10		10		10		3		5		3		5		5		10		10
750 to 1000	±0.66	±0.33	3	15	5	15	10	15	15	15	10	3	15	5	15	5	10	5	15	5	5	15	10	15
1000 to 1250	±0.76	±0.38		15		15		15		15		5		10		5		10		10		15		15
1250 to 1500	±0.82	±0.41		25		25		25		25		5		10		10		10		10		25		25
1500 to 1750	±0.86	±0.43		25		25		25		25		5		10		10		10		10		25		25
1750 to 2000	±0.92	±0.46		30		30		30		30		5		10		10		10		10		30		30

Belt Length		Length Tolerance	Inter-Shaft Distance Tolerance	2GT		3GT		EV5GT		EV8YU	
				Ci	Cs	Ci	Cs	Ci	Cs	Ci	Cs
	150 or Less	± 0.40	± 0.20		3		3		3		3
Over 150	250 or Less	± 0.40	± 0.20		3		3		3		3
Over 250	380 or Less	± 0.46	± 0.23		3		3		3		3
Over 380	500 or Less	± 0.50	± 0.25		3		3		3		3
Over 500	750 or Less	± 0.60	± 0.30	4	5	5	5	10	5	20	5
Over 750	1000 or Less	± 0.66	± 0.33		5		5		5		5
Over 1000	1250 or Less	± 0.76	± 0.38		10		10		10		10
Over 1250	1500 or Less	± 0.82	± 0.41		10		10		10		10
Over 1500	1750 or Less	± 0.86	± 0.43		10		10		10		10
Over 1750	2000 or Less	± 0.92	± 0.46		10		10		10		10

■Precautions on Operation

1. Be careful to avoid the ingress of foreign particles.
   When solid foreign particles enter during operation, it can scratch the belt and adversely affect the engagement of the belt and the pulley.
   In some cases, the pulley may disengage, land on the teeth of the pulley, and be cut.
2. Avoid Adhesion of oil.
   Oil on the rubber timing belt may wet and expand it, drastically shortening its service life.
   (a)Take special care when using solvent type oil.
   (b)A small amount of lubricant or grease, however, rarely causes a trouble.
3. Do not use the belt in a humid atmosphere.
4. Please use a well-ventilated safety cover.
5. The service life of the belt, when used at a high temperature (80˚C or more), can be drastically shortened.

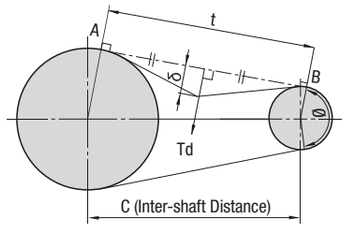
‹Reference›Belt Width Tolerance (Unit: mm)

Belt Width	Belt Length			
	351 or Less	351 to 840	840 to 1680	1680 or More
10 or Less	+0.3 −0.6	+0.3 −0.6	+0.3 −0.6	+0.6 −0.6
10 to 40	+0.6 −0.6	+0.6 −0.6	+0.6 −0.6	+0.6 −0.6
40 to 50	+0.6 −0.6	+0.6 −0.6	+1.0 −1.0	+1.0 −1.3

Cautions on Use of Belt

■How to Extend Belt

When the belt is too taut, its service life can be shortened, while when it is not taut enough, the belt may (jump off) the groove of the pulley due to an activating torque or shock load. Keep the belt stationary and optimize its tautness. The warp load necessary to provide the optimum tautness can be calculated from values representing the belt, its width and the span in equation A below. Apply deflection load between max. value and recommended value.



$$Td = \frac{Ti + \frac{t \times Y}{Lp}}{16} \quad \text{Equation A}$$

Td: Load N Needed for Deflection d at the Center of Span t

Ti : Initial Tension N — From Table 31
Y : Correction Coefficient — From Table 31
δ : Deflection (mm) — δ=0.016t
t : Span Length (mm) — $t = \sqrt{C^2 - \frac{(Dp-dp)^2}{4}}$
Lp : Length of the Belt (mm)
C : Inter-shaft Distance (mm)
dp : Diameter of the Pitch Circle of the Small Pulley (mm)
Dp : Diameter of the Pitch Circle of the Large Pulley (mm)

Table 32. Initial Tension (Ti) and Correction Coefficient (Y)

Type	Ti·Y	Belt Nominal Width / Belt Width mm	019 / 4.8	025 / 6.4	031 / 7.9	037 / 9.5	050 / 12.7	075 / 19.1	100 / 25.4	150 / 38.1	200 / 50.8
MXL	Ti (N)	Max. Value	9.8	13.7	–	21.6	29.9	–	–	–	–
		Recommended Value	5.8	8.2		12.9	18.0				
	Coefficient Y		–	–	–	–	–	–	–	–	–
XL	Ti (N)	Max. Value	–	29	37	44	67	–	–	–	–
		Recommended Value		18	25	32	51				
	Coefficient Y		–	3.8	5.4	7.6	11.8	–	–	–	–
L	Ti (N)	Max. Value	–	–	–	–	76	125	175	273	–
		Recommended Value					52	87	123	191	
	Coefficient Y		–	–	–	–	44.1	75.5	107	165	–
H	Ti (N)	Max. Value	–	–	–	–	–	293	421	646	889
		Recommended Value						222	312	486	668
	Coefficient Y		–	–	–	–	–	142	205	317	423

Type	Ti·Y	Belt Nominal Width / Belt Width mm	60 / 6	100 / 10	150 / 15	250 / 25
P2M	Ti (N)	Max. Value	13	–	–	–
		Recommended Value	9.8			
	Coefficient Y		0.9	–	–	–
P3M	Ti (N)	Max. Value	–	46	74	–
		Recommended Value		34	55	
	Coefficient Y		–	1.9	3.0	–
P5M	Ti (N)	Max. Value	–	147	225.4	–
		Recommended Value		107.8	166.6	
	Coefficient Y		–	56.9	82.4	–
P8M	Ti (N)	Max. Value	–	–	294	509.6
		Recommended Value			225.4	382.2
	Coefficient Y		–	–	135	239

Type	Ti·Y	Belt Nominal Width / Belt Width mm	40 / 4	60 / 6	100 / 10	150 / 15	250 / 25	300 / 30	400 / 40	600 / 60
S2M	Ti (N)	Max. Value	7.8	12.7	22.6	–	–	–	–	–
		Recommended Value	5.9	9.8	16.7					
	Coefficient Y		9.8	15.7	27.4	–	–	–	–	–
S3M	Ti (N)	Max. Value	–	26	46	73	–	–	–	–
		Recommended Value		20	34	54				
	Coefficient Y		–	26.5	46.1	75.5	–	–	–	–
S5M	Ti (N)	Max. Value	–	–	77	124	221	–	–	–
		Recommended Value			58	93	166			
	Coefficient Y		–	–	52.8	85.5	151.0	–	–	
S8M MTS8M	Ti (N)	Max. Value	–	–	–	294	510	628	873	
		Recommended Value				226	382	470	657	
	Coefficient Y		–	–	–	98	196	235	333	
S14M	Ti (N)	Max. Value	–	–	–	–	–	–	1226	1912
		Recommended Value							1108	1726
	Coefficient Y		–	–	–	–	–	–	686	1059

Type	Ti·Y	Belt Nominal Width / Belt Width mm	100 / 10	150 / 15	200 / 20	250 / 25	300 / 30	400 / 40	500 / 50
T5	Ti (N)	Max. Value	37.3	59	85	106	–	–	–
		Recommended Value	24.5	39	59	74			
	Coefficient Y		16.7	26.5	38.2	47.5	–	–	–
T10	Ti (N)	Max. Value	–	162	235	294	363	500	628
		Recommended Value		108	157	196	245	333	422
	Coefficient Y		–	71.6	104.9	130.4	163.8	222.6	281.5

Type	Ti·Y	Belt Nominal Width / Belt Width mm	4 / 4	6 / 6	9 / 9	12 / 12	15 / 15	20 / 20	25 / 25
2GT	Ti (N)	Max. Value	12.2	20.5	32.8	–	–	–	–
		Recommended Value	9.4	15.8	25.2				
	Coefficient Y		–	–	–	–	–	–	–
3GT	Ti (N)	Max. Value	–	38	57	–	96	–	–
		Recommended Value		29	44		74		
	Coefficient Y		–	–	–	–	–	–	–
EV5GT	Ti (N)	Max. Value	–	–	92	127	163	–	–
		Recommended Value			71	98	125		
	Coefficient Y		–	–	–	–	–	–	–
EV8YU	Ti (N)	Max. Value	–	–	–	–	273	364	455
		Recommended Value					210	280	350
	Coefficient Y		–	–	–	–	–	–	–